

## EXERCISE 12

- Write two Matlab functions:

- `[x,its] = bisection(f,a,b,tol,maxits)`, that implements the bisection method.
- `[x,its] = newton(f,derf,x0,tol,maxits)`, that implements the Newton method.

Implement suitable stopping criteria for the two methods. Use both functions to solve the equation  $e^x + \cos(x) - 2 = 0$ , whose exact solution is  $x = 0$  (choose appropriately the initial guess  $x_0 \neq 0$  or the initial interval). Plot the error  $|x_k - 0|$  vs  $k$ . Repeat the same for solving  $e^x - \cos(x) - 2 = 0$  whose solution is not known analytically and can be approximated by the previous methods with small tolerance or by `fsolve(@(x) exp(x)-cos(x)-2,0)` which calls the MATLAB solver.

- Use both methods to solve the equation  $\sqrt[3]{x - \pi} = 0$  (use the function `nthroot` to compute cubic roots). Plot the error  $|x_k - \pi|$  vs  $k$ . Discuss the results.